Outcome-Based Agents and Agentic Orchestration

Investment theses and category views on AI agents that complete enterprise work end to end and own the result

1. Investment theses: tailwinds, key takeaways and macro

This theme covers AI agents that complete a piece of enterprise work from start to finish, such as resolving a support ticket, working a denied medical claim or placing a purchase order, and that own the result, so the buyer receives finished work rather than a tool its staff operate. Vendors charge for that work per result, per unit, by usage or by subscription, so pricing is one way a vendor captures value and does not define the category. The theme also covers the orchestration software that lets a company run many agents under common rules. Five groups compete for this work, namely the model labs and cloud providers, the systems of record such as customer relationship management (CRM) and enterprise resource planning (ERP) platforms, enterprises building their own agents, systems integrators and outsourcers, and vertical applications built for one job in one industry.

1. **Agents can now finish whole jobs inside enterprise systems, so spending on them is moving from software budgets to the labor and outsourcing budgets that pay for completed work**
   - Model releases in the past year made agents reliable enough to run inside old enterprise software, and an insurance operations company says its success rate on tests that drive old insurance screens rose from 30% to over 95% with a single new model (company claim). Industry analysts expect agentic AI funding to grow 31.8% in 2027 against 3.7% for IT budgets overall (industry research).
   - The budget agents compete for is labor, which buyers measure in wages and outsourcing invoices. The US employs 2,666,000 customer service representatives at a median wage of $44,770 (BLS), about $119B a year by our estimate, and an outsourcing research firm expects $422B of business process outsourcing (BPO) spending to shift to software that does the work by 2035 (industry research).
   - Vendors are moving from tools that help staff to agents that do the job, and their pricing is starting to follow the work. The chief executive of a help desk platform says "seats are going to go away over time" (company interview), and the share of AI software vendors using outcome-based pricing rose from 18% to 23% in six months (investor survey).
   - The labs and cloud providers are paying to get agents installed, which shortens the path to production for every vendor. Partner funds of $100M and $750M now pay integrators to deploy agents (company announcements), and postings for forward-deployed engineers (FDEs), engineers who build each customer's deployment on site, passed 7,300 in the first nine months of 2026 against under 200 in 2023 (labor market data).
2. **New companies have an edge where they complete a whole job in one industry and own the result, because the labs sell general tools and the incumbents mostly add AI to the seats they already sell**
   - A vertical company knows the rules, exceptions and systems of one kind of work, which a general model does not learn from public data. A software chief executive argues that closing the gap between the roughly 70% accuracy of a model off the shelf and the 97% a business needs is what makes work automatable (podcast).
   - Getting an agent into production is mostly integration and testing, so a company that brings its own engineers and owns the finished work gets through where software alone stalls. A contact-center AI vendor's head of forward-deployed engineering says one month of building an agent can mean two to four months of testing and evaluation (expert call).
   - Companies that take legal or regulatory responsibility for the work keep more of the price, because buyers otherwise keep a person on the last step. An AI-native law firm whose lawyers take liability for the contracts its AI reviews sells the finished review of each contract to the client in place of software that a lawyer would operate (investor research).

- The incumbents often bolt agents onto old products and the labs avoid customer-specific work, which leaves room for specialists. A card issuer found the AI products of the contact-center incumbents "more of a bolt-on" (expert call), and a research agent founder says the labs avoid training on each customer's data because "they don't want to have a different model for every customer" (founder interview).

3. **Buyers judge an agent by the finished work it delivers against the outsourcer or headcount it replaces, and they want a predictable price more than a pure outcome contract**
   - The price an agent replaces is the outsourcer's invoice, and savings are wide enough to close deals quickly. A software company says offshore support cost "4X to 7X" what its AI agent costs (expert call).
   - Agent companies increasingly sell into the operating department's services budget, which is easier to reach than the IT software budget. An orchestration founder sells from the "service budget line, because it is easier to procure" (founder interview).
   - Buyers want to know what they will spend, so pure outcome pricing slows a sale even when the buyer likes the idea. In a fintech vendor selection, the buyer wanted "a more transactional pricing without outcome commitment" (expert call), and 27 of 50 buyers in an investor survey preferred credits tied to work over tokens (investor survey).
   - Buyers keep their data, procedures and tools in their own systems so any vendor can be replaced, which caps the price a vendor can hold once rivals catch up. A fintech routed tickets across three customer service vendors through a router it built itself (expert call).
4. **We should back agent companies whose completion rate and deployment cost already improve with each customer, because margins sit between outsourcing and software, and pricing per result is only one way to capture the value**
   - AI products earn gross margins of 50% to 60% against 80% to 90% for traditional software (investor research), and an investor survey projects AI product margins rising from 45% in 2025 to 59% in 2027 (investor survey). An insurance operations founder puts outsourcers at "10, 15 percent gross margin" and targets 80% for his own company (founder interview).
   - The share of work an agent finishes without a person drives margin, and per-resolution pricing makes the link direct because the vendor pays for model usage on every conversation but bills only the resolved ones. The finance chief of a customer service agent company says a resolution rate that rose from about 25% to close to 70% improved margin (company interview), and a rival runs about 90% of its workloads on cheaper fine-tuned open-source models (trade press).
   - Vendors that complete the same kind of work charge for it in different ways, so pricing follows the job and does not define it. Customer service platforms publish prices of $0.50 to $2.00 per resolution (vendor pricing pages), a customer service agent vendor that offers per-resolution pricing says most customers choose per conversation (company claim), an operations agent vendor charges one flat price per execution (founder interview), and a procurement vendor whose agents take requests from intake to purchase order sells an annual subscription with no share of savings (vendor pricing pages).
   - Deployment cost falls with each agent a customer adds, which is where margins improve over the life of an account. At a national retailer the first agent took six months and the second one month, and setup ran to "at least 35%-40% of your first year spend" (expert call).
   - A vendor paid a share of savings keeps a thin slice of the value it creates, so it needs volume to build a business. The same retailer measured a 35% to 40% cost cut and gave its vendor about 10% of it (expert call), so on $10M of handled cost the vendor earns $350,000 to $400,000 a year by our estimate.
5. **The labs and systems of record will absorb any agent that completes a generic task inside their own products, so we should back companies whose work needs industry knowledge or legal liability the platforms lack**
   - The labs now ship finished agents and give away the tools around them, which moves them into the job itself. A model lab now sells a complete customer service agent with written procedures, guardrails and

automated scoring of its answers (company announcement), and another has released plugins for 12 legal practice areas (company announcement).

- The systems of record are selling agents into their installed base with the customer data already in hand. The leading CRM vendor's agent product passed $1.5B of annual recurring revenue (ARR) on a definition widened to include its workplace chat assistant (company filings), and the leading IT service platform's AI products passed $1B of annual contract value (company filings).
- The main exit so far went to an incumbent at a price well below the private valuations of the leaders, which limits the return on late entries. A CRM incumbent paid about $3.6B for a leading customer service agent company (company announcement), while the two leading private rivals raised at roughly 75 to 100 times and about 45 times ARR by our estimate (company reports), with the lower multiple pairing a January 2026 valuation with revenue reported later in 2026. A company priced at 100 times ARR has to grow revenue about tenfold before a sale at 10 times ARR returns the entry price, by our estimate.
- Industry knowledge, a license and work that runs across many outside parties are the hardest things for a platform to copy, so new companies win in back-office and regulated work that enterprises already pay outsiders to do. New companies are concentrated there as well, with 87 of the 121 companies we track building vertical applications (our count).

2. Interesting categories

The categories follow the five groups in our primer, with vertical applications split into front office, back-office operations and regulated services, and each company is listed in one category only.

A. Labs and Hyperscalers

Agent Platforms, Frameworks and Infrastructure

- **Tailwinds: Agents are moving into production at most companies that build them, which turns agent building from an experiment into a recurring budget line.** In a survey of agent builders, 57% said they already had agents in production (developer survey).
- **Each major lab and cloud now sells a full platform to build, run and govern agents, so enterprise agent spending flows through a handful of suppliers.** The platforms launched in the past year add tools to build, deploy and manage agents, a managed agent loop and agent payments (company announcements).
- **Open-source agent frameworks have become a standard way for developers to build agents, and the clouds now plug them into their own deployment tools.** A cloud provider's tool now deploys, tests and monitors agents written in a leading open-source framework (developer documentation).
- **Every agent in production also needs testing, monitoring, permissions and billing, which adds spending around each agent a company runs.** In a reported incident, a coding agent deleted a production database and its backups in nine seconds using a token with broader rights than the task needed (press reports).
- **Headwinds: The labs give away the harness, the software that runs a model through a task and calls its tools, or price it near cost, which caps what an independent vendor can charge.** Lab agent APIs add no fee beyond model usage or charge about $0.08 per session-hour for a managed version (vendor pricing pages).
- **The labs are also building testing and improvement tools into their own agents, which removes the reason to buy them separately.** A lab-built customer service agent now ships with test conversations, automated scoring of answers and a loop that improves the agent, all of which independent testing vendors sell as products (company announcements).
- **Buyers rarely pay for neutrality across models, which weakens the main argument for an independent layer.** A managing director at a global integrator calls models "such commodities" and says routing between them is "hardly ever" used by his clients (expert call).
- **Why we should not focus here: The labs and hyperscalers own the models and agent platforms that every other category runs on, so in this theme they act as the suppliers and likely acquirers that**

companies in the other categories depend on. Independent frameworks and infrastructure sell to engineering teams on usage while the labs bundle the same tools free, so these companies compete with their suppliers and, by our estimate, tend to sell to a lab or cloud at a modest price.

Example companies: OpenAI, Anthropic, Google, Microsoft, Amazon Web Services, LangChain

B. Systems of Record

Incumbent Platforms as Channel, Acquirer and Absorber

- **Tailwinds: The record holders sit where the ticket, claim or order is opened and closed, so they can see and check every job an agent finishes.** Equity research says systems of record hold the customer data a result is measured against, which favors them as agents take on more work (equity research).
- **The record holders decide what counts as a finished job inside their own products, which lets them set the terms for every agent on their platform.** Help desk vendors decide for themselves when a resolution counts, either after their own model reviews the conversation or when the customer leaves without asking for more help, and cases handed to a person are not billed (vendor pricing pages).
- **The incumbents sell agents to customers they already have, which reaches buyers faster than any new vendor can.** Incumbent agent sales grow mainly inside the installed base, with half of agent bookings at the leading CRM vendor coming from existing customers buying more prepaid usage (company filings), and an advisory partner says the leading IT service platform's AI faces "few viable alternatives" in core IT service management (expert call).
- **The record holders are buying or backing agent companies built on their platforms, which gives those companies a likely acquirer.** In the past year incumbents in HR software, ERP and insurance core systems each bought or invested in an agent company that runs on their platform (company announcements).
- **Headwinds: Buyers distrust completion counts they cannot check, which slows every vendor that asks to be judged on finished work.** A software buyer audits its customer service vendor's resolution count itself because it cannot see how a resolution is defined (expert call), and a common default test treats 72 hours without a customer reply as a resolution (industry research).
- **Outside customer service, the incumbents mostly bill agents by the action or by prepaid usage, which keeps their revenue tied to software consumption more than to finished jobs.** The leading CRM vendor sells most agent work through prepaid credits at $0.10 per action (trade press), and the finance chief of an ERP vendor says revenue linked to outcomes will not move its 2027 numbers (company filings).
- **Seat-based incumbents lose seats as agents do the work, which pushes them to raise prices and leads buyers to question the platform.** An advisory partner reports shrinking seat counts and growth led by price increases at IT service platforms (expert call), and a retail chief information officer now asks whether he needs an IT service platform at all (expert call).
- **Why we should not focus here: Systems of record are established incumbents that most agent spending will run through, so they matter to us as the channel, the likely acquirer and the main absorber of any new company whose work they can copy.** They hold the data, permissions and audit trail an agent needs to finish work and prove it, so we should test every company in the theme against how hard its work is for the record holder to rebuild.

Example companies: Salesforce, Zendesk, HubSpot, ServiceNow, SAP, Oracle, Workday

C. Enterprises Building Their Own

Horizontal Orchestration and Work Context

- **Tailwinds: Enterprises run hundreds of systems and many agents, and someone has to coordinate them, set permissions and record what each agent did.** A founder in the category describes enterprises with "hundreds of applications" and thousands of people acting as glue between them (founder interview).

- **Agents perform far better when they know how work is actually done, which makes a record of real work patterns valuable to every agent a company runs.** A data governance chief executive cites agent accuracy rising from about 30% without business context to about 90% with it (trade press).
- **Coordination vendors report fast growth as enterprises add agents, and the integrators that deploy agents are starting to build practices around the layer.** Vendors in the layer report bookings above $150M and revenue growth above 300%, though the figures are their own (company claims), and one global integrator plans to put 1,000 engineers on a work-context platform (trade press).
- **Headwinds: Enterprises that buy a horizontal platform learn from it and build their next agents themselves, which caps how far each account grows.** Buyers use the platform to skip a long internal build and then take later use cases in-house, as one industrial buyer did after skipping a 12 to 14 month build, and it now expects its spending to grow only 5% to 10% (expert call).
- **Work context goes stale fast, so a map of how work gets done loses value unless someone keeps it current.** The same data governance chief executive says accuracy fell to 60% two months later (trade press).
- **The clouds and systems of record give the coordination layer away to win the platform.** An IT service platform made its agent control tower free for a year (company announcement), and a cloud provider now sells a product to govern every agent a company runs (company announcement).
- **The labs are moving into work context, so the companies that map work face the suppliers of the models they feed.** A director at a work-context vendor names the labs as his biggest competition and says customers will not need his company once the models learn to map work themselves (expert call).
- **Why we should not focus here: Horizontal orchestration sells a platform that enterprises increasingly build themselves or receive free from a platform they already own, so pricing power falls as the customer's own team learns.** Sales cycles for the biggest enterprise deals run about nine months by a founder's account (founder interview), and expansion stalls once the customer builds its own agents. Work context has the better economics of the two, because a record of how work really gets done improves every agent a company runs, but process-mining incumbents and the labs already compete for that layer, so we should not back standalone work-context companies. Most horizontal vendors now sell kits for building agents, which belong with the platforms in category A, so few companies remain in this category.

Example companies: Ema, Duvo, Convey, Workfabric AI

D. Systems Integrators and BPOs

AI Services and AI-Native Outsourcers

- **Tailwinds: Systems integrators are a main channel for enterprise agent deployments, and the labs are paying them to do the work.** Lab and cloud partner funds of $100M and $750M pay integrators for agent building, deployment and embedded engineers (company announcements).
- **Outsourcing is the budget agents replace, and the incumbent outsourcers are already losing ground.** Listed customer experience outsourcers report shrinking core revenue or write-downs on past acquisitions, including one write-down of $1.05B after a share price fall and a 1.0% fall in another's core services revenue in the second quarter of 2026 (company filings).
- **Automation lets services firms trade revenue for margin, which gives them a reason to adopt agents quickly.** A practice head at a listed IT services firm says the agent platform costs under $10,000 per worker a year against revenue above $20,000 per outsourced worker, and "Our revenues go down, but our margins go up" (expert call).
- **Headwinds: AI-run outsourcing is already in a price war, because the same models are available to every provider.** A vice president at a customer experience outsourcer says outsourcing rates for health insurers fell "between 20% and 40%" in two years and that "Right now, there is no protection in the industry" (expert call).

- **The customer keeps the data, so switching providers is easy and contracts do not compound.** An executive at an insurance company that buys outsourced operations work says "All the data is living within our existing systems" (expert call).
- **Discounting before the automation works erases the margin a services firm starts with.** Price cuts that run ahead of automation leave costs where they were, and at one AI-native outsourcer prices fell 10% to 40% on flat costs and gross margin "did tank heavily" (expert call).
- **Why we should not focus here: Services firms earn on people and hours, so automation shrinks their revenue even as it lifts their margin.** AI-native outsourcers that undercut the incumbents fight a price war with no switching costs because the data stays with the customer. Integrators matter to the theme as the channel that installs agents in enterprises, and the recurring value goes to the software they deploy, so we should back the vertical products that integrators resell and treat the services firms as partners and acquirers.

Example companies: Crescendo, Otera, Mechanical Orchard, Tessera Labs, Accenture, Deloitte, Cognizant, Teleperformance

E. Vertical Applications

E1. Front Office: Customer Service and Sales

- **Tailwinds: Customer service is the first job agents complete end to end at scale, because a resolved ticket is easy to define and check and the labor it replaces is easy to price.** Software companies spend $20 to $30 per ticket resolved by people, which gives buyers a clear benchmark for an agent that resolves the ticket itself (founder interview).
- **Savings at early buyers are wide enough to make the business case easy to defend.** Published and reported agent prices of about $0.50 to $2.00 per resolution sit far below the cost of a person, which runs $7.50 to $9 per transaction at one national retailer (vendor pricing pages, expert call).
- **Voice is opening a second wave of spending in contact centers that still run mostly on people.** The US still employs 2,666,000 customer service representatives (government data), and AI and self-service revenue at a listed contact-center software incumbent grew 52% to $362M of ARR (company filings).
- **Sales teams already pay agencies per meeting and per lead, which gives agents a known price to replace.** Agent vendors now bill sales work the same way, at published prices such as $9.99 per qualified lead, and another vendor bills only once the lead is verified in the customer's CRM or other connected systems (vendor pricing pages, trade press).
- **Headwinds: The labs, the help desk platforms and the contact-center incumbents all sell a customer service agent now, which pushes prices down for everyone.** A model lab launched its own customer service agent (company announcement), and a CRM incumbent now owns one of the leading agent vendors (company announcement).
- **Buyers spread work across vendors and consider building their own, so no vendor holds its account securely.** A fintech buyer says it would not bet on any single vendor and is considering building on local models of its own (expert call).
- **Results vary widely by customer, so buyers will soon compare agents with each other on resolution rate and price.** Resolution rates at one software customer sit at 30% to 40% against a reported 76% vendor average (expert call, company filings).
- **Sales agents churn heavily, because agents that send cold email burn through prospects and fail to show value inside short pilots.** An advisor to sales agent companies puts churn at 80% to 90%, the worst of any category he advises (expert call).
- **CRM incumbents bundle their own sales agents to replace point tools, which squeezes standalone vendors in their accounts.** One CRM bundle priced at $550 claims to replace roughly $1,300 of point-solution spend per US sales rep (company announcement).

- **Why we should not focus here: Customer service proves that agents can finish a whole job at scale, but the labs, the help desk platforms and the contact-center incumbents all sell the same finished job at falling prices.** Sales agents are paid on meetings and leads that are easy to count and easy to game, so buyers churn when quality slips, and the CRM owners bundle their own agents. Buyers in both functions keep their data so they can switch, and the customer service leaders already raised at roughly 45 to 100 times ARR by our estimate (company reports), which leaves a new entrant little room to win share or earn a strong return. We should use customer service resolution rates and prices as the benchmark for other functions and make no new bets in the category.

Example companies: Sierra, Decagon, Gradient Labs

E2. Back-Office Operations: Finance, Compliance and Supply Chain

- **Tailwinds: Finance teams repeat the same work every month, and each invoice, collection or reconciliation has a clear count and cost.** Because each unit is already counted, buyers can judge an agent on each finished invoice or collection and check the result in their own books, and vendors in payables and collections report volumes above 100M invoices a year and overdue balances cut by 40% (company claims).
- **Procurement savings are measured against total spend, which ties an agent's value to the money it saves on purchases as well as the labor it replaces.** For a manufacturer spending $10B a year, a 50% more efficient team saves $50M, while a 1% to 2% cut in purchase prices is worth $100M to $200M (founder interview), and a procurement agent vendor says it charges only on realized savings (company claim).
- **Procurement and finance agents run on top of the ERP without replacing it, which speeds approval and lets the buyer check results in its own records.** At one global retailer the agent runs beside the existing ERP and already handles 45% to 50% of procurement volume (expert calls).
- **Accounting and payables agents now sell the finished job and take responsibility for it.** An accounting agent vendor says it never had a seat or token price for its core product, and a payables vendor says its agents decide whether to pay each invoice and "take accountability for the outcome" (company interviews).
- **Compliance alerts and freight claims arrive as countable cases, much like invoices, and outsourcers and automation vendors now sell agents that clear them.** A compliance agent vendor says alerts pile into backlogs that regulators fine when hiring lags volume (company site), an automation vendor sells agents built for financial crime work (earnings call), and an outsourcer with its agent partner packages anti-money laundering checks and freight claims as agent solutions (trade press).
- **Headwinds: ERP vendors are building agents into the systems many finance teams already run, which limits how far a new vendor can reach inside those accounts.** A leading ERP vendor sells its agents through paid AI units, with a rules layer validating every action the model proposes (trade press).
- **Buyers keep a person on the final step, so the full price of the job moves to the agent slowly.** The chief executive of a procurement outsourcer says "I don't have any client that lets it go on its own" (expert call).
- **Procurement agents that run beside the buyer's record are easy to replace, which limits how long a vendor holds an account.** One global retailer that pays on a consumption license reports little stickiness and puts a replacement of its procurement agent vendor at seven to eight months (expert call).
- **Why we should focus here: Finance, compliance and supply chain work repeats every month, is easy to count and costs money when it goes wrong, so buyers pay for agents that cut the cost of each invoice, investigation, claim or order.** The agents sit on the buyer's ERP, so the result is counted in a record the buyer already trusts, a simple procurement deployment can be set up in a few days (expert call), and nobody has to replace a system to start. The work also runs across suppliers, carriers, banks and email outside any single record, which makes it harder for an ERP vendor to absorb than a help desk task. Revenue recurs with the monthly close and grows with the spend the agent handles, and we should back companies whose cost per invoice or order already falls with each new customer, because that is when margins move toward software levels.

Example companies: Basis, Freehand, Magentic, Didero, Salient, Bretton AI, BackOps, Niural

E3. Regulated Services: Healthcare, Insurance, Legal and Life Sciences

- **Tailwinds: Healthcare billing, insurer approval before treatment and referrals are rule-bound work that providers and health plans pay for by the claim or by the dollar recovered.** Fees to health plans for prior authorization can include a charge per automatically approved request (trade press), and at least one coding review vendor bills hospitals only for findings that are approved and billed (vendor pricing pages).
- **Insurance runs on outsourced servicing, underwriting support and claims work, much of it on old systems that agents can now operate directly.** Agents can now drive old insurance screens reliably enough for production (company claim), and one insurer reports its servicing agent resolves more than 90% of service cases without a person (company claim).
- **Companies that hold a license or act as the service provider can sell the whole job and keep its economics.** An AI-native commercial brokerage and AI-native claims administrators now sell placement and claims handling, from first notice to payout, as a service (trade press).
- **Legal agents are paid from the legal services budget that pays outside counsel, often per matter.** Personal injury demand packages, the file of evidence and damages a lawyer sends an insurer to settle a claim, are now priced per package (investor research), and an AI-native consumer law firm reports revenue above $35M (trade press).
- **Drug companies are starting to buy research agents, beginning with target discovery, finding the gene or protein a new drug should act on, which gives vendors a paying customer for a clear unit of work.** A research agent vendor says one run equals about 6 months of a scientist's work (company claim), and a listed drug developer is deploying a research agent across its research work (trade press).
- **Headwinds: The dominant electronic health record vendor controls the record at most big health systems and limits outside AI, which slows new vendors in those accounts.** A revenue cycle executive says the vendor does not "allow integrations of AI outside of what they're already working on" (expert call).
- **Providers still review claims before submission because errors carry liability, so the agent does not yet replace the whole job.** A dermatology group's chief information officer asks, "If any of the wrong claims go in, who takes a liability?" (expert call).
- **The labs are moving into healthcare administration with their own tools and sample workflows.** A model lab released healthcare connectors and a sample skill for reviewing requests for insurer approval before treatment (company announcement).
- **Insurance buyers are pitched constantly and expect cheaper offers built on open models, which caps prices for vendors that sell tools.** An insurance executive says he is pitched "at least two or three a day" and expects a rival offer "at a 10th of the price by using open-source models" (expert call).
- **Why we should focus here: Regulated services are bought by the claim, the policy, the matter or the run, so an agent that completes the unit replaces work the buyer already buys from outsourcers and law firms.** Companies that hold a license or take legal responsibility for the result keep more of the price than tool vendors and face the least pressure from cheaper open models. Healthcare billing agents are paid on cash the provider sees in its own books, and the record vendor's limits matter least in independent and specialty practices. As drafting tools get cheap, we should back the firms that sell finished legal work under their own license, because they keep the price of the outside counsel they replace. Research outcomes take years to prove, and a research agent priced at $200 per run ended its founding subscriptions seven months after launch (company announcements), so in life sciences we should back companies that already show repeat paid runs at the same customer.

Example companies: EvenUp, Crosby, Lawhive, Anterior, SmarterDx, Procode, TJM Labs, Adaptive Innovations, Alaffia Health, Reserv, Harper, Pace, Strala, Edison Scientific